MIL-E-20090A
24 October 1967

SUPERSEDING
MIL-P-20090
6 September 1951
(See 6.8)

MILITARY SPECIFICATION

ENAMEL, ALKYD, STRIPING, BROWN (FORMULA NO. 41)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers enamel, alkyd, striping, brown, Formula No. 41, for use as a marking paint with good brushing characteristics.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-D-643	- Drier, Paint, Naphthenate, Liquid, Concentrated.
TT-P-143	- Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.
TT-P-405	- Pigment, Metallic-Brown, Dry.
TT-P-463	- Pigment; Zinc Oxide, Dry and Paste-in-Oil.
TT-R-266	- Resin, Alkyd, Solutions.
TT-T-291	- Thinner; Paint, Volatile Spirits (Petroleum-Spirits).

MILITARY

MIL-P-15173 - Pigment, Magnesium Silicate, Dry (Paint Pigment).

STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials; Method of Inspection, Sampling, and Testing.

MILITARY

MIL-STD-755 - Labels Containing Symbols For Packages And Containers For Hazardous Industrial Chemicals And Materials.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Formula.

3.1.1 The enamel shall consist of ingredients conforming to the requirements of the applicable specifications in the proportions specified in table I. Small amounts of antiskinning agents, suspension agents, and antidrier adsorption agents may be added to assist in meeting the requirements of this specification and the amount of thinner may be adjusted slightly to meet the viscosity requirement provided that all the other requirements of the specification are met and the exact formula used is furnished in the record required by method 1031 of FED-STD-141.

3.1.2 The formula specified in table I is designated standard Formula No. 41. Whenever Formula No. 41 is specified, the paint shall meet the requirements of this specification.

Table I - Formula No. 41

Ingredients	Pounds 1/
Metallic brown (TT-P-405)	280
Zinc oxide (type I, grades A or B of TT-P-463)	40
Magnesium Silicate (MIL-P-15173)	100
Alkyd resin solution (Type I, Class A, TT-R-266)(see 6.6)	465
Paint thinner (Grade 1, TT-T-291)	195
Lead naphthenate drier (Type I, TT-D-643)	16.4
Cobalt naphthenate drier (Type II, TT-D-643)	3.3

1/ The formula is given slightly in excess of 100 gallons to allow for normal manufacturing loss (see 6.7).

3.2 Manufacture. The component raw materials shall be mixed and dispersed as required to produce a product which is uniform, homogeneous and entirely suitable for the purpose intended, and in full conformity to the requirements of this specification.

3.3 Quantitative requirements. The enamel shall conform to the quantitative requirements of table II, when tested in accordance with 4.4.

Table II - Quantitative requirements.

Characteristics	Requirements	
	Minimum	Maximum
Pigment, percent by weight of enamel	36.5	40.5
Volatiles, percent by weight of enamel	28.5	32.5
Nonvolatile vehicle, percent by weight of enamel (Calculated by difference)	29.0	33.0
Phthalic anhydride, percent by weight of nonvolatile vehicle	23.0	- -
Water, percent by weight of enamel	- -	0.5
Coarse particles and skins, as residue retained on No. 325 sieve, percent by weight of pigment	- -	0.2
Consistency, Krebs units, within 30 days of manufacturing	72.	82.
Weight per gallon, pounds	10.4	11.0
Drying time		
Set-to-touch, hours	- -	2.0
Time of drying hard, hours	- -	8.0
Gloss, 60° specular	30	- -
Flash point, °F.	100	- -
Fineness of grind	4	- -
Zinc oxide, percent by weight of pigment	8.0	11.0
Iron oxide (Fe_2O_3), percent by weight of pigment	44.5	- -

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3.4 Qualitative requirements. The enamel shall conform to the following qualitative requirements.

3.4.1 Odor. The odor shall be normal for the volatiles permitted when tested as specified in table III.

3.4.2 Condition in container. A freshly opened full container of enamel shall be free from grit, seeds, skins, lumps, abnormal thickening, and livering, and shall show no more pigment settling or caking than can be readily reincorporated to a smooth homogeneous state.

3.4.3 Storage stability.

3.4.3.1 Partially full container. A three-quarter filled, closed 8-ounce glass jar of the enamel shall show no skinning at the end of 48 hours when tested as specified in 4.4.4.1. After aging as specified in 4.4.4.1, the enamel shall show no livering, curdling, seeding, hard caking, or gummy sediment. It shall mix readily to a smooth homogeneous state, and any skin formed shall be continuous and easily removed.

3.4.3.2 Full container. The enamel, when tested as specified in 4.4.4.2, shall show no skinning, livering, curdling, seeding, hard caking, nor tough gummy sediment, shall remix readily to a smooth homogeneous state, shall have a maximum viscosity of 93 Krebs units, and shall meet all other requirements of the specification for at least one year after date of manufacture.

3.4.4 Brushing properties. The enamel, when tested as specified in 4.4.5, shall be capable of being brushed out and laid off without excess drag on the brush. When dry, the brushed surface shall be free from sags and runs and shall show a minimum of brush marks.

3.4.5 Spraying properties. The enamel, when tested as specified in 4.4.6, shall show no running, sagging, or streaking. The film shall show no dusting, mottling, or color separation.

3.4.6 Flexibility. A film of the enamel, prepared and tested as specified in 4.4.7 shall withstand bending without cracking or flaking.

3.4.7 Water resistance. A film of the enamel, prepared and tested as specified in 4.4.8 shall show no wrinkling or blistering when examined immediately after removal from distilled water. When examined 2 hours after removal, there shall be no excessive softening, whitening, or dulling. After 24 hours air drying, the portion of the panel which was immersed shall be almost indistinguishable from the portion of the panel that was not immersed with regard to hardness, adhesion, and general appearance.

3.4.8 Color. The color shall match the dry standard color card (see 6.4) when tested as specified in 4.4.9.

3.4.9 Dilution stability. When thinned as specified in 4.4.10, the enamel shall remain stable and uniform and shall show no precipitation, curdling, or separation. Slight pigment settling shall be permitted.

3.4.10 Toxicity. The enamel shall contain no benzene (benzol), chlorinated compounds, or hydrolyzable chlorine derivatives, when inspected in accordance with table III.

3.4.11 Rosin. The nonvolatile vehicle shall not show the presence of rosin when tested as specified in table III.

3.4.12 Phenolic resin. The nonvolatile vehicle shall not show the presence of phenols when tested as specified in table III.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality assurance. Quality assurance shall be provided in accordance with method 1031 of FED-STD-141 as hereinafter supplemented. Submission of form in accordance with method 1031 shall cover complete tests other than the full container storage stability test. The full container storage stability requirements shall apply for one year after manufacture regardless of other testing or prior acceptance of the material.

4.3 Additional inspection.

4.3.1 Ingredient materials. When requested by the testing laboratory or other controlling authority, one pint of each ingredient in the formula specified in table I shall be supplied for test purposes.

4.4 Test procedures. The enamel shall be tested in accordance with the applicable methods of FED-STD-141 and compliance to the requirements specified in table III.

Table III - Index

Item	Test Method		Section of this Specification giving Requirements
	Applicable method in FED-STD-141	Section of this specification giving further reference	
Pigment content	4021	-----	Table II
Volatiles	4041	-----	Table II
Nonvolatile vehicle content	4053	-----	Table II
Phthalic anhydride	7021	-----	Table II
Water	4081	-----	Table II
Coarse particles	4091	-----	Table II
Consistency, Krebs-Stormer	4281	-----	Table II
Weight per gallon	4184	-----	Table II
Drying time	4061	4.4.1	Table II
Specular gloss	6101	4.4.2	Table II
Fineness of grind	4411	-----	Table II
Flash point	4293	-----	Table II
Zinc oxide	7291	-----	Table II
Iron oxide	7161	-----	Table II
Odor	4401	-----	3.4.1
Condition in container	3011	4.4.3	3.4.2
Stability in partially full container	3021	4.4.4.1	3.4.3.1
Stability in full container	3022	4.4.4.2	3.4.3.2
Brushing properties	4321	4.4.5	3.4.4
Spraying properties	4331	4.4.6	3.4.5
Flexibility	6221	4.4.7	3.4.6
Water resistance	6011	4.4.8	3.4.7
Color	4250	4.4.9	3.4.8
Dilution stability	4203	4.4.10	3.4.9
Toxicity	----	4.4.11	3.4.10
Resin	5031	-----	3.4.11
Phenolic resins	5141	-----	3.4.12

4.4.1 Drying time. Determine drying time in accordance with method 4061 of FED-STD-141, except that the laboratory conditions shall be as specified in Section 7 of FED-STD-141.

4.2.2 Specular gloss. Determine gloss in accordance with method 6101 of FED-STD-141, except that the laboratory conditions shall be as specified in Section 7 of FED-STD-141.

4.4.3 Condition in container. The supplier shall determine package condition at time of acceptance in accordance with method 3011 of FED-STD-141 and observe for compliance to 3.4.2.

4.4.4 Storage stability.

4.4.4.1 Partially full container. Determine skinning after 48 hours in accordance with method 3021 of FED-STD-141. Reseal and age for seven days at 60°C. and observe for compliance to 3.4.3.1.

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4.4.4.2 Full container. The Government, at its option and at any time not to exceed one year after manufacture, may test enamel stored in its original containers for package condition, viscosity, dry hard time, and color difference from the standard color card in accordance with method 3022 of FED-STD-141. (Any action by the Government to reject enamel after prior acceptance shall be based on the examination of enamel stored in its original containers.) For the purpose of developing a quality history of the supplier's product, the Government shall allow a full one-quart can of enamel to stand undisturbed at a temperature of 21°C to 32°C, 70°F to 90°F) for a period not to exceed 12 months and then shall examine for compliance to 3.4.3.2.

4.4.5 Brushing properties. Apply the enamel as packaged in accordance with method 4321 of FED-STD-141 and observe for compliance to 3.4.4.

4.4.6 Spraying properties. Reduce five parts by volume of paint with one part by volume of thinner conforming to grade 1 of TT-T-291. Spray on a steel panel to give a dry film thickness between 0.0009-0.0011 inch and observe for spraying properties in accordance with method 4331 of FED-STD-141 for compliance to 3.4.5. For referee test, use automatic application as specified in method 2131 of FED-STD-141.

4.4.7 Flexibility. Determine flexibility in accordance with method 6221 of FED-STD-141. Apply a 2 inch wide film of enamel with a film applicator that will give a dry film thickness of 0.0009 to 0.0011 inch on a flat tin plate panel prepared as in method 2012 of FED-STD-141 using the petroleum naphtha-ethylene glycol monoethyl ether mixture. Air dry the enamel in a horizontal position for 2 hours, than bake for 24 hours at 105° + 2°C. (221° + 4°F.). Condition at 23° + 1°C for 1/2 hours and bend over a 1/8-inch mandrel. Examine in a strong light with 7-diameter magnification for compliance to 3.4.6.

4.4.8 Water resistance. Draw down a film of the enamel with a 0.002-inch (0.004-inch gap clearance) film applicator on a flat tin plate panel prepared as in method 2012 of FED-STD-141 using the petroleum naphtha-ethylene glycol monoethyl ether mixture. Air dry the enamel for 48 hours and then immerse in distilled water at 23° + 1°C. for 18 hours in accordance with method 6011 of FED-STD-141. At the end of the test period remove and allow to recover for 2 hours. Examine for compliance to 3.4.7.

4.4.9 Color. In accordance with method 4250 of FED-STD-141, compare the standard color card (see 6.4) with the film of enamel on white glass panel prepared for the gloss test (see 4.4.2) and observe for compliance to 3.4.8. If doubt exists as to the acceptability of the match, determine the color difference by instrument as specified in method 6123 of FED-STD-141. An acceptable color match shall be exact or within two units in the direction of minus Δ L, Δ a, and Δ b.

4.4.10 Dilution stability. Reduce one part by volume of enamel as packaged with one part by volume of mineral spirits conforming to grade 1 of TT-T-291. Then test according to method 4203 of FED-STD-141 for compliance to 3.4.9.

4.4.11 Toxicity. The manufacturer shall certify that the enamel contains no benzene (benzol), chlorinated compounds, or hydrolyzable chlorine derivatives.

4.5 Packaging, packing, and marking. The enamel shall be inspected for compliance with the packaging, packing, and marking requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. The enamel shall be packaged and packed in accordance with TT-P-143. The level of packaging shall be level A or C, and the level of packing shall be level A, B, or C, as specified (see 6.2). The size and type of container shall be as specified (see 6.2).

5.2 Marking. The containers shall be marked in accordance with TT-P-143. In addition to any special marking specified in the contract or order, each container shall have affixed a warning label of appropriate size similar to Class 2 of MIL-STD-755, or shall be lithographed or stencilled with a reasonable likeness thereof. Under "Contains" shall be inserted "petroleum thinners of 100°F minimum flash point." For unit containers that also serve as shipping containers, any conflict with ICC regulations shall be resolved by reasonable modification of size of label or use of warning statement without label design.

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6. NOTES

6.1 Intended use. Brown striping enamel is used as a marking paint for applications not requiring a fast dry. The enamel has good brushing properties, high hiding power, and strong tinting strength.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and level of packing required (see 5.1).
(c) Size and type of container required (see 5.1).
(d) Special marking required (see 5.2).

6.3 Paint should be purchased under this specification by volume, the unit being one U. S. gallon of 231 cubic inches at 15.5°C (60°F).

6.4 Color cards. Color cards may be obtained upon application to the Chemical Laboratory, Norfolk Naval Shipyard, Portsmouth, Virginia 23709. The purpose for which the cards are desired should be specified. These cards are the only standards acceptable for use for the requirements of this specification in inspection testing and as such should be used by the manufacturer in color matching his product. For rough approximations of the color not involving the color matching by the manufacturer or inspection testing, color 20061 of FED-STD-595 may be used.

6.5 Verification inspection by the Government will be limited to the amount deemed necessary to determine compliance with the contract or order, and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract or order. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the supplier's quality control system and the quality history of the product.

6.6 If it is desired to use alkyd resin solution conforming to the requirements for type I, class B of TT-R-266, the weight of the alkyd resin solution, as specified in this specification, should be multiplied by 1.17 and the paint thinner, petroleum spirits (Grade I of TT-T-291), reduced by 0.17 times the weight of alkyd resin solution specified in this specification.

6.7 Composition by volume. For information only, and with the understanding that weight-volume relationships of ingredients may vary slightly, the following approximation of composition by volume is included.

Ingredients	Gallons	
Metallic brown	8.21	
Zinc oxide	0.86	
Magnesium silicate	4.21	
Alkyd resin solution	58.10	(37.30)[1/]
Paint thinner	29.50	
Lead naphthenate drier	1.71	
Cobalt naphthenate drier	0.41	
Total volume	103.00	

[1/] Figure in parentheses refers to the volume of resin solid (nonvolatile).

6.8 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Custodians:
Army - MR
Navy - SH

Review activities:
Army - MR
Navy - SH

User activities:
Navy - AS, OS

Preparing activity:
Navy - SH
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE
(NAVSHIPS OVPRT 12-66)

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